

Recovering the initial values from the solutions of the fractional wave equation on the light cone trace

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Introduction

light cone

- (a) The path in all directions through which the light can travel along space-time.
- (b) Satisfies the wave equation because light is a wave.
- (c) Can express light cone as $t = \pm|\mathbf{x}|$ and here we deal with $t = |\mathbf{x}|$ case.

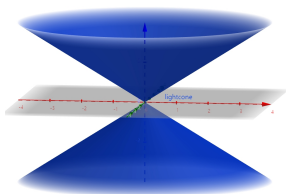


Figure: light cone.

- The wave equation

$$\begin{aligned} \partial_t^2 p(\mathbf{x}, t) &= \Delta_{\mathbf{x}} p(\mathbf{x}, t), \quad p(\mathbf{x}, 0) = f(\mathbf{x}), \quad \text{and } D_t p(\mathbf{x}, t)|_{t=0} = 0 \\ (\mathbf{x}, t) &= (x_1, x_2, t) \in \mathbb{R}^2 \times (0, \infty) \end{aligned} \quad (1)$$

$$p(\mathbf{x}, t) = \frac{1}{(2\pi)^2} \int_{\mathbb{R}^2} e^{i\xi \cdot \mathbf{x}} \cos(|\xi|t) \mathcal{F}f(\xi) d\xi, \quad (2)$$

- Define the linear operator $\mathcal{W}_2$

$$\mathcal{W}_2 f(\mathbf{x}) = p(\mathbf{x}, |\mathbf{x}|). \quad (3)$$

Goal : reconstruction $f(\mathbf{x})$ from $p(\mathbf{x}, |\mathbf{x}|)$

- Rakesh and Yuan provided reconstruction of the initial function ONLY IN ODD dimensions using Radon transform in 2018.

Strategy

$$\mathcal{W}_2 f \longmapsto (\mathcal{W}_2 f)_m \longmapsto (\mathcal{F}f)_m \longmapsto \mathcal{F}f \longmapsto f$$

- The fractional wave equation

$$\begin{aligned} D_t^\alpha p(\mathbf{x}, t; \alpha) &= -(-\Delta_{\mathbf{x}})^{\frac{\alpha}{2}} p(\mathbf{x}, t; \alpha), \\ (\mathbf{x}, t) &= (x_1, \dots, x_n, t) \in \mathbb{R}^n \times (0, \infty), \quad 1 < \alpha \leq 2 \end{aligned} \quad (4)$$